

# Towards Ultrasound-Based Blood Pressure Monitoring: A Hybrid Array Enabling Continuous Chirp Excitation for Increased Axial Resolution

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## Background, Motivation and Objective

Continuous non-invasive blood pressure (BP) monitoring remains a major challenge in clinical diagnostics. Cuff-based techniques provide only intermittent measurements and limited comfort, while invasive arterial lines enable continuous BP assessment but carry a significant infection risk. Novel photoplethysmography (PPG) based approaches allow continuous non-invasive monitoring but rely on indirect estimation via light absorption, requiring frequent calibration and providing limited accuracy. In contrast, ultrasound-based BP estimation via arterial diameter tracking provides a direct alternative but requires high axial resolution in combination with sufficient signal to noise ratio. To address this, we propose a hybrid transducer array combining broadband piezoelectric transmission with capacitive micromachined ultrasound transducers (CMUTs), enabling continuous chirp excitation for high-resolution vascular monitoring.

## Statement of Contribution/Methods

Our miniaturized hybrid acoustic stack integrates two lead magnesium niobate–lead titanate (PMN-PT) single crystal piezoelectric transmitters with five CMUT receiving channels (a), supported by a custom analog front end. For excitation, we used a broadband chirp signal to enhance axial resolution. To evaluate the system's ability to capture arterial wall motion, we utilized a porcine artery embedded in an IEC 60601-2-37-compliant agar-based tissue-mimicking model (b). Pressure was varied in the phantom while M-mode imaging captured wall motion. Performance comparison was conducted between short pulse excitation (c) and the proposed broadband chirp method (d).

## Results/Discussion

The acoustic stack supports successfully arterial wall imaging and enables precise arterial diameter measurement. Compared with conventional short-pulse excitation, continuous chirp excitation achieved approximately 33 dB higher amplitude, and 2.75 times reduced full width at half maximum (FWHM). Continuous chirp excitation, enabled by the separate transmit and receive channels, significantly improved image quality and resolution during simultaneous operation. These results demonstrate the feasibility and advantages of the hybrid approach for vascular blood pressure monitoring.

